AF-3403 **BCA1C1**

B.C.A. DEGREE EXAMINATION, APRIL 2011

First Semester

Computer Applications

PROGRAMMING IN C

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a variable ?

2. Write down the purpose of ‘ Goto ’ statement.

3. How can you initialize an array ?

4. What is the difference between ‘strcpy’ and ‘strncpy’ functions ?

5. Write any *two* Library functions.

6. Give an example for array of structures.

7. Define a Pointer.

8. Which is a generic pointer ? Give its Syntax.

9. How can you name a file ?

10. What is a Macro ?

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Describe the structures of 'C' programs.

(*Or*)

(b) Discuss about 'switch' statement.

12. (a) Write a program to transpose a given matrix.

(*Or*)

(b) Discuss about any *three* string handling functions.

13. (a) Write briefly about nesting of functions.

(*Or*)

(b) Differentiate Structures with Unions.

14. (a) How can you declare and initialize pointers ?

(*Or*)

(b) Discuss about Pointers and Arrays.

15. (a) Write short notes an Command line arguments.

(*Or*)

(b) How can you include a file ? Explain.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe the various data types in C.

17. How strings can be read and written ? Discuss with examples.

18. Discuss about the various categories of functions.

19. Write a program using pointers to read an array of integers and print the same.

20. Discuss about the I/O operations on files.

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AF-3404 **BCA2C1**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Second Semester

Computer Applications

PROGRAMMING IN C++

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write any *four* features of Object oriented programming.

2. What is meant by overloading of a function ? When do you use this concept ?

3. When do we declare a member of a class static ?

4. What is the purpose of destructors ?

5. What is an abstract class ?

6. What role does the iomanip file play ?

7. What are the advantages of saving data in binary form ?

8. Write down overloading resolution steps when template functions are overloaded.

9. What are Exceptions ? What are the types of exception ?

10. What is unit testing ?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain the new ( ) operator.

(*Or*)

(b) Explain Default arguments.

12. (a) With an example, explain how objects can be passed as function arguments.

(*Or*)

(b) Write a program to construct a matrix of size $m \times n$.

13. (a) How is multipath inheritance handled in C++ ?

(*Or*)

(b) How are constructors defined in derived classes ? What is the sequence of executing the constructors ?

14. (a) Write a program to handle Disk file using Class objects.

(*Or*)

(b) Write the syntax of function template. Use this concept to Swap two data.

15. (a) Explain Fault Tolerant Design technique.

(*Or*)

(b) Explain the steps in Object oriented analysis.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Write a program to calculate the Variance and Standard Deviation of N numbers.

17. Explain various types of Constructors.

18. Explain Virtual function with an example.

19. Explain handling Uncaught exceptions.

20. Write a program to overload operators in complex class. Use template class.

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AF-3405 **BCA3C1**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Third Semester

Computer Applications

DATABASE MANAGEMENT SYSTEMS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by weak and strong entity sets.

2. What is meant by normalization ?

3. What is canonical cover ?

4. What is temporal data ?

5. State the reasons for building distributed systems.

6. What is SAN ?

7. Define Foreign key.

8. Define Object privileges.

9. State the Trigger components.

10. State the benefits of stored procedures.

Part B (5 × 5 = 25)

Answer **all** the questions choosing **either** (a) **or** (b).

11. (a) Explain briefly about the extended E-R features.

(*Or*)

(b) Discuss about the design phases of a database.

12. (a) Discuss in detail about the server system architectures.

(*Or*)

(b) Discuss briefly on modeling temporal data.

13. (a) Discuss about the issues in studying parallelism and also about the factors against efficient parallel operations.

(*Or*)

(b) Discuss about the partitioning techniques in detail.

14. (a) Explain the process of creating, renaming and removing synonyms.

(*Or*)

(b) Discuss briefly about indexes and sequences.

15. (a) Explain the package structure with example.

(*Or*)

(b) Explain how to create, execute and remove a stored procedure.

Part C (3 × 10 = 30)

Answer any **three** of the following.

16. Explain in detail about data models and database languages.

17. Explain decomposition using multivalued dependencies.

18. Discuss briefly about Data replication and Data fragmentation.

19. Discuss in detail about user privileges and roles.

20. Explain in detail about Triggers and Sensors.

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AF-3406 BCA4C1

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fourth Semester

Computer Applications

JAVA PROGRAMMING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is polymorphism ?

2. What is JVM ?

3. Give the general form of conditional operator.

4. What is the difference between while and do-while statement ?

5. Define Constructors.

6. What is the difference between string and string buffer classes ?

7. What is packages in Java ?

8. List any two common run time errors.

9. Write the minimum parameters of applet tag.

10. What is canvas ?

Part B

$(5 \times 5 = 25)$

Answer **all** the questions choosing **either** (a) **or** (b).

11. (a) Write a note on JDK.

(*Or*)

(b) Write note on Java tokens.

12. (a) Discuss about Type conversions.

(*Or*)

(b) List and explain decisions making statements in Java.

13. (a) Explain inheritance in detail.

(*Or*)

(b) Explain wrapper classes in detail.

14. (a) How will you implement multiple inheritance in Java ? Explain with example.

(*Or*)

(b) Write a note on thread priority and synchronization.

15. (a) Explain about a web page design.

(*Or*)

(b) How will you pass parameters to applets ? Explain with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the basic concepts of OOPs with examples.

17. What are the different types of operators in Java ? Explain each of them with examples.

18. Explain about vector classes and its methods in detail.

19. Explain the exception handling mechanism and common Java exceptions in detail.

20. Explain applet life cycle with neat diagram.

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AF-3407

BCA5C1

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

MULTIMEDIA

(CBCS—2008 onwards)

Time : 3 Hours

Maximum : 75 Marks

Part A

(10 × 2 = 20)

Answer **all** questions.

1. What is the use of view menu ?

2. What is Stage ?

3. List any two basic Object Selection tools.

4. What is Current Frame ?

5. How to use text in a Shape Tween ?

6. What is Morph ?

7. List the Command which is used to undo all the changes made to a file.

8. What is the use of Resample up option ?

9. Define Layer.

10. Define Filters.

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) Explain the Workspace.

(*Or*)

(b) Explain Flash Default File type.

12. (a) Differentiate Solid colors with Gradient colors.

(*Or*)

(b) How to create background with frames ? Explain.

13. (a) Motion Tween playback is very jumpy-not smooth. Why ?

(*Or*)

(b) What causes the dotted line to appear in the green interpolated frames of shape tween ? Explain.

14. (a) Explain the function of Option bar and Navigator palette.

(*Or*)

(b) How to fill a selection with a pattern in photoshop ? Explain.

15. (a) Explain Layer effects.

(*Or*)

(b) How to convert Point type to Paragraph type ? Explain.

Part C (3 × 10 = 30)

Answer any **three** out of five questions.

16. Flow to transform drawn objects. Explain in detail.

17. Explain in detail Reshaping objects with lines and fills.

18. How to add sounds and sound effects to an animation ? Explain.

19. Explain the selection tools in photoshop.

20. Explain the concept of working with Layers.

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AF-3408 BCA5C2

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

COMPUTER SYSTEM ARCHITECTURE AND DESIGN

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Convert the decimal number 41.6875 into binary.

2. What is bus ?

3. What do you mean by interrupt ?

4. What is the use of Compiler ?

5. What are the most common fields found in instruction format ?

6. State the purpose of Zero flag.

7. Classify the I/O commands.

8. What is Priority Interrupt ?

9. Define Hit ratio.

10. Define Address space.

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the different types of Complements.

(*Or*)

(b) What are the two types of control organization ? Explain them.

12. (a) Draw the flowchart for interrupt cycle and explain.

(*Or*)

(b) How can you write a program in assembly language ? Also compare with machine language.

13. (a) How the stack is useful to evaluate the expression ?

(*Or*)

(b) List out the program control instructions and state the purpose of each.

14. (a) Describe the algorithm for adding and subtracting numbers in signed 2's complement representation.

(*Or*)

(b) With an example, explain I/O interface.

15. (a) Write about memory hierarchy.

(*Or*)

(b) What is page replacement and explain any two page replacement algorithms ?

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. What are the basic instruction formats available in a computer and explain them ?

17. Describe the major tasks that must be performed by the assembler during the translation process.

18. Describe the different types of addressing modes.

19. Explain the DMA transfer with an example.

20. Explain the organization of Associative memory.

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AF-3409 **BCAE1A**

B.C.A DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

Elective—DATA MINING AND DATA WAREHOUSING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by query management process ?

2. Define Data warehousing.

3. Name three different Data warehouse process managers.

4. What is meant by Data life expectancy ?

5. What is the use of sequential discovery ?

6. What is meant by Data mining ?

7. What are two types of machine learning ?

8. What is the use of decision tree ?

9. Define Large itemset.

10. What is meant by support ?

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe the method of directing Queries.

(*Or*)

(b) Describe the architecture of Load Manager.

12. (a) What are the three main features required for the management of backups ? Explain.

(*Or*)

(b) Describe the method of tuning the data load.

13. (a) Write about Clustering and Classification.

(*Or*)

(b) Explain social implications of Data mining.

14. (a) Describe the use of indexing in multi dimensional data

(*Or*)

(b) What are two commonly used similarity measures in traditional IR systems ? Descrribe each one with formula.

15. (a) Write the sampling algorithm.

(*Or*)

(b) Write short notes on ‘ Large itemsets ’.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the architecture of a Data warehouse.

17. Explain the responsibilities of the Warehouse Manager.

18. Explain the five steps in KDD process.

19. Discuss fuzzy sets and fuzzy logic in detail.

20. Explain parallel and distributed algorithms.

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AF-3410 **BCAE1B**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

Elective : WEB DESIGN TECHNOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What is the use of <HR> tag ?

2. Describe the use of SRC attribute in IMG element.

3. Differentiate Break and Continue statement.

4. What is meant by looping ?

5. Define Function.

6. What is meant by scope of a variable ?

7. What are collections ?

8. What is meant by object referencing ?

9. What is TDC ?

10. What is the use of scale method ?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the use of Headers with examples.

(*Or*)

(b) Name and explain the Syntax of a tag used to create link in HTML page.

12. (a) Describe the syntax of if selection structure in javascript.

(*Or*)

(b) What are different logical operators used in javascript ? Explain.

13. (a) Describe the method of declaring and allocating arrays in Javascript.

(*Or*)

(b) Write a script that inputs text from an HTML form and outputs the text in uppercase letters.

14. (a) Describe the method of creating style sheets with the STYLE element.

(*Or*)

(b) Write a script to trigger an ONCLICK event.

15. (a) Describe the use of direct animation path control.

(*Or*)

(b) Explain the method of creating image masks.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the method of creating ordered and unordered lists in a HTML web page.

17. Explain the syntax of for structure in javascript with an example.

18. Write a script to print histogram.

19. Discuss the usage of any two filters in DHTML.

20. Write a DHTML script to demonstrate background audio with BGSOUND.

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AF-3411 **BCAE2A**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

Elective : WIRELESS APPLICATION PROTOCOL

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define WAP. Mention the components of WAP Standard.

2. What is a history stack ?

3. What is the purpose of Cache value ?

4. Write short notes on URL.

5. Write about any two public WTAI functions.

6. What is a Cryptolibrary ?

7. What is a Transcoder ?

8. Mention the methods to reduce the keystrokes while designing an user interface.

9. Write about the headers defined by CC/PP Exchange Protocol.

10. What is a push message ?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write short notes on Service Protocols of WAP Standard.

(*Or*)

(b) Explain about WAP deployment chain.

12. (a) Write about the basics of WML.

(*Or*)

(b) Write short notes on variables in WML.

13. (a) Write about Binary WML Script.

(*Or*)

(b) What are the functions in WML Script standard library regarding 'Math' and 'Number' ?

14. (a) Mention the application design goals of mobile Internet.

(*Or*)

(b) Explain about user profile designing.

15. (a) What are the techniques using HTTP 1.1 to customize content for the client ?

(*Or*)

(b) Write about WTA Security Model.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Give an overview of WAP Architecture.

17. Explain in detail about any one control element in WML.

18. Write about the operators supported by WML script.

19. Write the general design guidelines for designing an user Interface.

20. Explain about WTA Application Programming Interface.

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AF-3412 **BCAE2B**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Fifth Semester

Computer Applications

Elective : ADVANCED JAVA PROGRAMMING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is RMI ?

2. Name the *four* abstract classes of Stream based I/O.

3. What is an event ? Give an example.

4. How is add ( ) method called in Japplet ?

5. What is modeless dialog box ?

6. What is the purpose of Layout Manager ?

7. What is Proxy Server ?

8. What is DNS ?

9. What is Introspection ?

10. What is a Cookie ?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain Input stream class.

(*Or*)

(b) What is Pushback Reader ? Explain its methods.

12. (a) Explain Event Classes.

(*Or*)

(b) Explain Inner Classes.

13. (a) How are Check Boxes handled in Java ?

(*Or*)

(b) Explain the methods and classes used for creating a hierarchy of nested sub menus.

14. (a) Explain URL Connection Class.

(*Or*)

(b) What is Datagram ? Explain Datagram Packet Class.

15. (a) Discuss the design patterns for properties of a Bean's State.

(*Or*)

(b) What is a Session ? How is Session Tracked ?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the classes that support character streams.

17. Explain the Delegation event model to handle keyboard events. Write code wherever necessary.

18. Explain the creation and usage of Trees in an applet. Write code wherever necessary.

19. Explain TCP/IP Client Sockets. Write a suitable program to show the usage.

20. What is a Servlet ? Explain the life cycle of a Servlet.

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AF-3413 **BCA6C1**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Sixth Semester

Computer Applications

COMPUTER GRAPHICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Write any two applications of Image processing.

2. How can you load an intensity value into the frame buffer at a position (x, y) ?

3. Define Scaling.

4. Specify the Concatenation Properties.

5. Define Aspect ratio.

6. Which is called a window ?

7. What do you mean by shearing ?

8. Define Uniform scaling in 3D transformations.

9. List down any *two* components of user interface.

10. What is a feedback ?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Describe any one of the input device.

(*Or*)

(b) Explain Bresenham's algorithm for drawing a line.

12. (a) Discuss about any one of the basic transformations.

(*Or*)

(b) Discuss about the matrix representations of basic transformations.

13. (a) Write short notes on viewing transformations.

(*Or*)

(b) Write briefly about line clipping.

14. (a) Describe three - dimensional Rotation.

(*Or*)

(b) Write down the matrix representations of scaling about the origin and about a selected fixed print.

15. (a) Write briefly about the User's model.

(*Or*)

(b) What is Information Display ? Explain.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about the application of Computer Graphics.

17. Discuss about the Composite transformations.

18. Describe Convex polygon clipping with an example.

19. Explain the following :

(a) Translation.

(b) Composite transformations.

20. Discuss about the Command language and its styles.

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AF-3414

BCA6C2

B.C.A. DEGREE EXAMINATION, APRIL 2011

Sixth Semester

Computer Applications

COMPUTER NETWORKS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Give any two design issues for the Layers.

2. What is Bluetooth ?

3. Describe the frame format for Bit oriented protocol.

4. Write about Parity bit.

5. What is congestion ?

6. What is fragmentation ?

7. What is Nagle's Algorithm ?

8. Mention the types of Multiplexing.

9. What is IMAP ?

10. What is web mail ?

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Write short notes on various types of Transmission media.

(*Or*)

(b) How Computer network is useful for home appliances ?

12. (a) Write about error detection and correction codes.

(*Or*)

(b) What is the role of Finite State Machine model in protocol verification ?

13. (a) Discuss about the general principles of Congestion Control.

(*Or*)

(b) Discuss about Tunneling.

14. (a) Write about Transport service primitives.

(*Or*)

(b) Write about Flow control and Buffering.

15. (a) Explain about Client and Server Side of www.

(*Or*)

(b) How did you compress an audio file ?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about OSI reference model with neat diagram.

17. Write about 1–bit sliding window protocol.

18. Explain about any two Adaptive routing algorithm.

19. Discuss about Network performance measurement.

20. Explain about any two methods of Cryptography.

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AF-3415 **BCA6C3**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Sixth Semester

Computer Applications

CLIENT / SERVER COMPUTING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Client / Server Computing.

2. Expand OLTP.

3. What is a GUI ?

4. Abbreviate DLL.

5. What is multithreading ?

6. Expand DRDA.

7. What is the function of Network layer ?

8. List out the five ways to interconnect LAN.

9. What are the MIS services carried over to LAN environment ?

10. List out any two Network operating system.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) List out the benefits of Client / Server computing

(*Or*)

(b) Discuss the components of Client / Server Applications.

12. (a) Write a note on Client operating system.

(*Or*)

(b) Explain the basic principles of effective GUI design.

13. (a) Explain how RPC works.

(*Or*)

(b) Write a note on Trigger.

14. (a) Explain TCP / IP Model.

(*Or*)

(b) List out the advantages and disadvantages of Bridges.

15. (a) Write a note on Reliability and Serviceability.

(*Or*)

(b) Discuss about Database administrator training.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the categories of Client / Server Applications.

17. Discuss about DLL, DDE and OLE.

18. Explain the different categories of servers.

19. Explain OSI model with block diagram.

20. Discuss the issues of LAN and Network management.

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AF-3416 BCAE3A

B.C.A. DEGREE EXAMINATION, APRIL 2011

Sixth Semester

Computer Applications

Elective—SOFTWARE ENGINEERING

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Software Engineering.

2. Write any four factors that influence the quality and productivity.

3. Name the two categories of Cost estimation techniques.

4. What are the characteristics of software ?

5. List the Coupling Techniques.

6. Write Bohm-Jacopini theorem.

7. Distinguish between Verification and Validation.

8. Write a standard and guideline for software maintenance.

9. How can you calculate an error Index during the development of software ?

10. List out Cost activities for quality.

Part B (5 × 5 = 25)

Answer **all** the questions.

11. (a) Discuss briefly on the managerial issues of Software Engineering.

(*Or*)

(b) Write short notes on MBO.

12. (a) Write a note on Delphi cost estimation.

(*Or*)

(b) Discuss briefly on Decision table.

13. (a) Write a note on recursion in software.

(*Or*)

(b) Explain briefly on Standards and Guidelines.

14. (a) Write short notes on Validation Testing.

(*Or*)

(b) Discuss on "Debugging is an art".

15. (a) Describe Software quality activities.

(*Or*)

(b) Write short notes on Quality.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain Software Life Cycle Model.

17. Elucidate various software cost factors.

18. Discuss about Design Notations.

19. Explain Integration Testing.

20. Describe the Formal Technical Reviews.

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AF-3417 **BCAE3B**

B.C.A. DEGREE EXAMINATION, APRIL 2011

Sixth Semester

Computer Applications

Elective—MOBILE COMMUNICATION

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. What are the layers in a wireless and mobile environment ?

2. Define "Coherence time".

3. What is polling ?

4. Write any two applications of Satellite system.

5. Mention any two basic mechanisms for MAC layer.

6. Write the function of Radio Access layer.

7. Write the advantages and disadvantages of Transaction Oriented TCP.

8. Give a mechanism of Transmission time out freezing.

9. What is push technology ?

10. Mention any two Wireless telephony applications.

Part B $(5 \times 5 = 25)$

Answer **all** the questions.

11. (a) What is multiplexing ? Also write a note on CDM.

(*Or*)

(b) Write a note on Path loss of radio signals.

12. (a) Discuss briefly on various Aloha.

(*Or*)

(b) Write short notes on DECT system architecture.

13. (a) Distinguish Infra-red and Radio transmission.

(*Or*)

(b) Write a note on Addressing.

14. (a) Write an alternative metrices of mobile ad-hoc networks.

(*Or*)

(b) Discuss briefly on mobile TCP.

15. (a) Write a note on HTTP.

(*Or*)

(b) Write short notes on Wireless markup language.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain Cellular system.

17. Describe GSM logical channels and Frame hierarchy.

18. Explain WATM services and its Reference model.

19. Compare Indirect TCP with Snooping TCP.

20. Explain wireless transaction protocol.

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